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URL of this page: https://medlineplus.gov/asthma.html

Asthma

What is asthma?

Asthma is a chronic (long-term) lung disease. It affects your airways, the tubes that carry air in and out of your lungs. When you have asthma, your airways can become inflamed and narrowed. This can cause wheezing, coughing [https://medlineplus.gov/cough.html] , and tightness in your chest. When these symptoms get worse than usual, it is called an asthma attack or flare-up.

What causes asthma?

The exact cause of asthma is unknown. Genetics and your environment likely play a role in who gets asthma.

An asthma attack can happen when you are exposed to an asthma trigger. An asthma trigger is something that can set off or worsen your asthma symptoms. Different triggers can cause different types of asthma:

- Allergic asthma is caused by allergens. Allergens are substances that cause an allergic reaction. They can include
  - Dust mites
  - Mold [https://medlineplus.gov/molds.html]
  - Pets
  - Pollen from grass, trees, and weeds
  - Waste from pests such as cockroaches and mice
- Nonallergic asthma is caused by triggers that are not allergens, such as
  - Breathing in cold air
  - Certain medicines
  - Household chemicals [https://medlineplus.gov/householdproducts.html]
  - Infections such as colds [https://medlineplus.gov/commoncold.html] and the flu [https://medlineplus.gov/flu.html]
  - Outdoor air pollution [https://medlineplus.gov/airpollution.html]
  - Tobacco smoke [https://medlineplus.gov/secondhandsmoke.html]
- Occupational asthma is caused by breathing in chemicals or industrial dusts at work
- Exercise-induced asthma happens during physical exercise, especially when the air is dry

Asthma triggers may be different for each person and can change over time.

Who is at risk for asthma?

Asthma affects people of all ages, but it often starts during childhood. Certain factors can raise your risk of having asthma:

- **Being exposed to secondhand smoke** when your mother is pregnant with you or when you are a small child
- **Being exposed to certain substances at work**, such as chemical irritants or industrial dusts
- **Genetics and family history.** You are more likely to have asthma if one of your parents has it, especially if it's your mother.
- **Race or ethnicity.** Black and African Americans and Puerto Ricans are at higher risk of asthma than people of other races or ethnicities.
- **Having other diseases or conditions** such as obesity [https://medlineplus.gov/obesity.html] and allergies [https://medlineplus.gov/allergy.html]
- **Often having viral respiratory infections** as a young child
- **Sex.** In children, asthma is more common in boys. In teens and adults, it is more common in women.

What are the symptoms of asthma?

The symptoms of asthma include:

- Chest tightness
- Coughing, especially at night or early morning
- Shortness of breath [https://medlineplus.gov/breathingproblems.html]
- Wheezing, which causes a whistling sound when you breathe out

These symptoms can range from mild to severe. You may have them every day or only once in a while.

When you are having an asthma attack, your symptoms get much worse. The attacks may come on gradually or suddenly. Sometimes they can be life-threatening. They are more common in people who have severe asthma. If you are having asthma attacks, you may need a change in your treatment.

How is asthma diagnosed?

Your health care provider may use many tools to diagnose asthma:

- Physical exam
- Medical history
- Lung function tests [https://medlineplus.gov/lab-tests/lung-function-tests/] , including spirometry, to test how well your lungs work
- Tests to measure how your airways react to specific exposures. During this test, you inhale different concentrations of allergens or medicines that may tighten the muscles in your airways. Spirometry is done before and after the test.
- Peak expiratory flow (PEF) tests to measure how fast you can blow air out using maximum effort
- Fractional exhaled nitric oxide (FeNO) tests to measure levels of nitric oxide in your breath when you breathe out. High levels of nitric oxide may mean that your lungs are inflamed.
- Allergy skin [https://medlineplus.gov/lab-tests/allergy-skin-test/] or blood [https://medlineplus.gov/lab-tests/allergy-blood-test/] tests, if you have a history of allergies. These tests check which allergens cause a reaction from your immune system.

What are the treatments for asthma?

If you have asthma, you will work with your health care provider to create a treatment plan. The plan will include ways to manage your asthma symptoms and prevent asthma attacks. It will include:

- **Strategies to avoid triggers.** For example, if tobacco smoke is a trigger for you, you should not smoke or allow other people to smoke in your home or car.

- **Short-term relief medicines,** also called quick-relief medicines. They help prevent symptoms or relieve symptoms during an asthma attack. They include an inhaler to carry with you all the time. It may also include other types of medicines which work quickly to help open your airways.
- **Control medicines.** You take them every day to help prevent symptoms. They work by reducing airway inflammation and preventing narrowing of the airways.

If you have a severe attack and the short-term relief medicines do not work, you will need emergency care.

Your provider may adjust your treatment until asthma symptoms are controlled.

Sometimes asthma is severe and cannot be controlled with other treatments. If you are an adult with uncontrolled asthma, in some cases your provider might suggest bronchial thermoplasty. This is a procedure that uses heat to shrink the smooth muscle in the lungs. Shrinking the muscle reduces your airway's ability to tighten and allows you to breathe more easily. The procedure has some risks, so it's important to discuss them with your provider.

Start Here

- About Asthma [https://www.cdc.gov/asthma/about/] (Centers for Disease Control and Prevention)
  Also in Spanish [https://www.cdc.gov/asthma/es/about/acerca-del-asma.html]
- Asthma [https://www.niehs.nih.gov/health/topics/conditions/asthma]
  (National Institute of Environmental Health Sciences)
- What Is Asthma? [https://www.epa.gov/asthma/what-asthma] (Environmental Protection Agency)
- What Is Asthma? [https://www.nhlbi.nih.gov/health/asthma] (National Heart, Lung, and Blood Institute)
  Also in Spanish [https://www.nhlbi.nih.gov/es/salud/asma]

Diagnosis and Tests

- Cough [https://familydoctor.org/symptom/cough/] (American Academy of Family Physicians)
- Lung Function Tests [https://medlineplus.gov/lab-tests/lung-function-tests/] (National Library of Medicine)
  Also in Spanish [https://medlineplus.gov/spanish/pruebas-de-laboratorio/pruebas-de-funcion-pulmonar/]
- Spirometry [https://www.lung.org/lung-health-diseases/lung-procedures-and-tests/spirometry]
  (American Lung Association)
- Spirometry [https://www.mayoclinic.org/tests-procedures/spirometry/about/pac-20385201?p=1]
  (Mayo Foundation for Medical Education and Research)
  Also in Spanish [https://www.mayoclinic.org/es/tests-procedures/spirometry/about/pac-20385201?p=1]
- What is Fractional Exhaled Nitric Oxide (FeNO) Testing?
  [https://www.nhlbi.nih.gov/sites/default/files/publications/FeNO-Testing.pdf]
  (National Heart, Lung, and Blood Institute) - **PDF**

Prevention and Risk Factors

- Allergens and Irritants [https://www.niehs.nih.gov/health/topics/agents/allergens]
  (National Institute of Environmental Health Sciences)
- Asthma Triggers: Gain Control [https://www.epa.gov/asthma/asthma-triggers-gain-control]
  (Environmental Protection Agency)
  Also in Spanish [https://espanol.epa.gov/cai/tome-control-de-los-factores-desencadenantes-de-su-asma]
- Controlling Asthma [https://www.cdc.gov/asthma/control/] (Centers for Disease Control and Prevention)
  Also in Spanish [https://www.cdc.gov/asthma/es/control/como-controlar-el-asma.html]

- Reducing Allergens in Your Home [https://www.nhlbi.nih.gov/sites/default/files/publications/Reducing-Allergens_rev_508.pdf] (National Heart, Lung, and Blood Institute) - **PDF**

Treatments and Therapies

- Asthma and Complementary Health Approaches: What You Need to Know [https://www.nccih.nih.gov/health/asthma-and-complementary-health-approaches-what-you-need-to-know] (National Center for Complementary and Integrative Health)
- Asthma Management Guidelines and Your Care [https://www.nhlbi.nih.gov/sites/default/files/publications/Asthma-Management-Guidelines-Your-Care_rev_508.pdf] (National Heart, Lung, and Blood Institute) - **PDF**
- Can Immunotherapy Help with the Treatment of Allergic Asthma? [https://www.nhlbi.nih.gov/sites/default/files/publications/Immunotherapy-Allergic-Asthma.pdf] (National Heart, Lung, and Blood Institute) - **PDF**
- Changing Role of Inhaled Corticosteroids in Asthma Management [https://www.nhlbi.nih.gov/sites/default/files/publications/Inhaled-Corticosteroids.pdf] (National Heart, Lung, and Blood Institute) - **PDF**
- Inhaled Asthma Medications [https://www.aaaai.org/tools-for-the-public/conditions-library/asthma/inhaled-asthma-medications] (American Academy of Allergy, Asthma, and Immunology)
- Is Bronchial Thermoplasty Right for You? [https://www.nhlbi.nih.gov/sites/default/files/publications/Bronchial-Thermoplasty.pdf] (National Heart, Lung, and Blood Institute) - **PDF**
- Long-Acting Muscarinic Antagonists (LAMAs) [https://www.nhlbi.nih.gov/sites/default/files/publications/Long-Acting-Muscarinic-Antagonists-LAMA.pdf] (National Heart, Lung, and Blood Institute) - **PDF**
- Long-Term Control Medications for Lung Diseases [https://www.nationaljewish.org/conditions/medications/asthma-medications/longterm?modal=1] (National Jewish Health)
- Manage Your Asthma: Know Your Triggers and Treatment Options [https://www.fda.gov/consumers/consumer-updates/manage-your-asthma-know-your-triggers-and-treatment-options] (Food and Drug Administration)
  Also in Spanish [https://www.fda.gov/consumers/articulos-para-el-consumidor-en-espanol/controle-su-asma-conozca-los-detonantes-y-sus-opciones-de-tratamiento]
- Quick-Relief Medications for Lung Disease Symptoms [https://www.nationaljewish.org/conditions/medications/asthma-medications/quickrelief?modal=1] (National Jewish Health)
- Suicide Warnings on Medications [https://aapp.org/faq/471018/suicide-warnings-medications] (American Association of Psychiatric Pharmacists) - **PDF**
- Understand Your Asthma Medication [https://www.lung.org/lung-health-diseases/lung-disease-lookup/asthma/treatment/medication] (American Lung Association)
- What's the Difference Between a Nebulizer and an Inhaler? [https://kidshealth.org/en/teens/nebulizer-inhaler.html] (Nemours Foundation)

Living With

- Asthma Action Plan [https://www.nhlbi.nih.gov/sites/default/files/publications/Asthma-Action-Plan-2020_rev_508.pdf] (National Heart, Lung, and Blood Institute) - **PDF**
- Asthma Diet: Does What You Eat Make a Difference? [https://www.mayoclinic.org/diseases-conditions/asthma/expert-answers/asthma-diet/faq-20058105?p=1] (Mayo Foundation for Medical Education and Research)
  Also in Spanish [https://www.mayoclinic.org/es/diseases-conditions/asthma/expert-answers/asthma-diet/faq-20058105?p=1]

- Asthma Inhalers: Which One's Right for You? [https://www.mayoclinic.org/diseases-conditions/asthma/in-depth/asthma-inhalers/art-20046382?p=1] (Mayo Foundation for Medical Education and Research)
- Measuring Your Peak Flow Rate [https://www.lung.org/lung-health-diseases/lung-disease-lookup/asthma/treatment/devices/peak-flow] (American Lung Association)
- Monitoring Your Asthma [https://www.nhlbi.nih.gov/resources/lmbb-monitoring-your-asthma-fact-sheet] (National Heart, Lung, and Blood Institute) - **PDF**
- So You Have Asthma: A Guide for Patients and Their Families [https://www.nhlbi.nih.gov/resources/so-you-have-asthma-guide-patients-and-their-families] (National Heart, Lung, and Blood Institute) - **PDF**

Related Issues

- Allergies and Asthma: They Often Occur Together [https://www.mayoclinic.org/diseases-conditions/asthma/in-depth/allergies-and-asthma/art-20047458?p=1] (Mayo Foundation for Medical Education and Research)
- Asthma and COPD: Differences and Similarities [https://www.aaaai.org/tools-for-the-public/conditions-library/asthma/asthma-and-copd-differences-and-similarities] (American Academy of Allergy, Asthma, and Immunology)
- Can Asthma Be Predicted? Early Signs, Risk Factors & Diagnosis in Children [https://www.healthychildren.org/English/health-issues/conditions/allergies-asthma/Pages/Asthma-Predictive-Index.aspx]
- Flu and People with Asthma [https://www.cdc.gov/flu/highrisk/asthma.html] (Centers for Disease Control and Prevention)
- Gastroesophageal Reflux Disease [https://www.aaaai.org/conditions-treatments/related-conditions/gastroesophageal-reflux-disease] (American Academy of Allergy, Asthma, and Immunology)
  Also in Spanish [https://www.aaaai.org/conditions-treatments/afecciones-y-tratamientos/afecciones-relacionadas/enfermedad-de-reflujo-gastroesofagico-(erg)]
- Guide to Controlling Asthma at Work [https://www.lung.org/lung-health-diseases/lung-disease-lookup/asthma/managing-asthma/workplace] (American Lung Association)
- Humidifiers: Ease Skin, Breathing Symptoms [https://www.mayoclinic.org/diseases-conditions/common-cold/in-depth/humidifiers/art-20048021?p=1] (Mayo Foundation for Medical Education and Research)
- People at Higher Risk of Flu Complications [https://www.cdc.gov/flu/highrisk/] (Centers for Disease Control and Prevention)
- Respiratory Infections and Asthma [https://www.cdc.gov/asthma/respiratory-infections/] (Centers for Disease Control and Prevention)
- Sleep Problems in Asthma and COPD [https://www.thoracic.org/patients/patient-resources/resources/sleep-problems-asthma-copd.pdf] (American Thoracic Society) - **PDF**
  Also in Spanish [https://www.thoracic.org/patients/patient-resources/resources/spanish/sleep-problems-in-asthma-and-copd.pdf]
- Vocal Cord Dysfunction: Is it a Type of Asthma? [https://www.mayoclinic.org/diseases-conditions/asthma/expert-answers/vocal-cord-dysfunction/faq-20058019?p=1] (Mayo Foundation for Medical Education and Research)
  Also in Spanish [https://www.mayoclinic.org/es/diseases-conditions/asthma/expert-answers/vocal-cord-dysfunction/faq-20058019?p=1]

Specifics

- Asthma Attack [https://www.mayoclinic.org/diseases-conditions/asthma-attack/symptoms-causes/syc-20354268?p=1] (Mayo Foundation for Medical Education and Research)
  Also in Spanish [https://www.mayoclinic.org/es/diseases-conditions/asthma-attack/symptoms-causes/syc-20354268?p=1]
- Exercise and Asthma [https://www.aaaai.org/tools-for-the-public/conditions-library/asthma/exercise-and-asthma] (American Academy of Allergy, Asthma, and Immunology)

Also in Spanish [https://www.aaaai.org/tools-for-the-public/biblioteca-de-condiciones/biblioteca-de-asma/ejercicio-y-asma]

- Exercise-Induced Asthma [https://www.mayoclinic.org/diseases-conditions/exercise-induced-asthma/symptoms-causes/syc-20372300?p=1] (Mayo Foundation for Medical Education and Research)
- Exercise-Induced Bronchoconstriction Defined [https://www.aaaai.org/tools-for-the-public/allergy,-asthma-immunology-glossary/exercise-induced-bronchoconstriction-defined] (American Academy of Allergy, Asthma, and Immunology)
- Exercise-Induced Bronchospasm [https://familydoctor.org/condition/exercise-induced-bronchospasm/] (American Academy of Family Physicians)
- FAQs About Steroids for Asthma [https://www.nationaljewish.org/conditions/medications/asthma-medications/longterm/faqs-about-steroids-for-asthma?modal=1] (National Jewish Health)
- Nocturnal Asthma [https://www.nationaljewish.org/conditions/asthma/overview/types/nocturnal-asthma?modal=1] (National Jewish Health)

Genetics

- Allergic asthma: MedlinePlus Genetics [https://medlineplus.gov/genetics/condition/allergic-asthma] (National Library of Medicine)

Images

- Metered dose inhaler use - Series [https://medlineplus.gov/ency/presentations/100200_1.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/esp_presentations/100200_1.htm]
- Nebulizer use - series [https://medlineplus.gov/ency/presentations/100201_1.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/esp_presentations/100201_1.htm]
- Peak flow meter use - Series [https://medlineplus.gov/ency/presentations/100202_1.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/esp_presentations/100202_1.htm]
- Spacer use - Series [https://medlineplus.gov/ency/presentations/100203_1.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/esp_presentations/100203_1.htm]

Videos and Tutorials

- Self-Care for Asthma-Using Your Inhaler [https://www.cdc.gov/asthma/caring/] (Centers for Disease Control and Prevention)

Test Your Knowledge

- Asthma Quiz [https://www.aaaai.org/conditions-treatments/asthma/asthma-quiz] (American Academy of Allergy, Asthma, and Immunology)
- Asthma Treatment Quiz [https://medlineplus.gov/ency/quiz/000141_36.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/quiz/000141_116.htm]
- EIB (Exercise-Induced Bronchoconstriction) Asthma Quiz [https://www.aaaai.org/conditions-treatments/asthma/eib-quiz] (American Academy of Allergy, Asthma, and Immunology)
- Test Your Knowledge About Inhalers [https://medlineplus.gov/ency/quiz/000005_32.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/quiz/000005_115.htm]

Statistics and Research

- Inflammation [https://www.niehs.nih.gov/health/topics/conditions/inflammation] (National Institute of Environmental Health Sciences)

Journal Articles

References and abstracts from MEDLINE/PubMed (National Library of Medicine)

- Article: Effect of combined strength and endurance training in adults with asthma:... [https://www.ncbi.nlm.nih.gov/pubmed/41370562]
- Article: Aspirin Dosing for Secondary Prevention of Atherosclerotic Cardiovascular Disease in Chronic... [https://www.ncbi.nlm.nih.gov/pubmed/41294093]
- Article: A cluster randomized controlled trial of an electronic medical record-based pathway... [https://www.ncbi.nlm.nih.gov/pubmed/41285460]
- Asthma -- see more articles [https://pubmed.ncbi.nlm.nih.gov/?term=%22Asthma%22%5Bmajr%3Anoexp%5D+AND+humans%5Bmh%5D+AND+english%5Bla%5D+AND+%22last+1+Year%22+%5Bedat%5D+AND+%28patient+education+handout%5Bpt%5D+OR+guideline%5Bpt%5D+OR+clinical+trial%5Bpt%5D%29++NOT+%28letter%5Bpt%5D+OR+case+reports%5Bpt%5D+OR+editorial%5Bpt%5D+OR+comment%5Bpt%5D%29+AND+free+full+text%5Bsb%5D+]

Find an Expert

- American Academy of Allergy, Asthma, and Immunology [https://www.aaaai.org/]
- American Lung Association [https://www.lung.org/]
- Find an Allergist/Immunologist [https://allergist.aaaai.org/find/]
  (American Academy of Allergy, Asthma, and Immunology)
- National Heart, Lung, and Blood Institute [https://www.nhlbi.nih.gov/] NIH
- National Institute of Allergy and Infectious Diseases [https://www.niaid.nih.gov/] NIH

Children

- Asthma in Children: MedlinePlus Health Topic [https://medlineplus.gov/asthmainchildren.html]
  NIH (National Library of Medicine)
  Also in Spanish [https://medlineplus.gov/spanish/asthmainchildren.html]

Women

- Asthma and Pregnancy [https://mothertobaby.org/fact-sheets/asthma-and-pregnancy/]
  (Organization of Teratology Information Specialists)
  Also in Spanish [https://mothertobaby.org/es/hojas-informativas/asma-embarazo/]
- Asthma, Allergies and Pregnancy [https://www.aaaai.org/tools-for-the-public/conditions-library/asthma/asthma,-allergies-and-pregnancy-ttr] (American Academy of Allergy, Asthma, and Immunology)
  Also in Spanish [https://www.aaaai.org/tools-for-the-public/biblioteca-de-condiciones/biblioteca-de-asma/asma,-alergias-y-embarazo-consejos-para-tener-en-c]

Older Adults

- Medications and Older Adults [https://www.aaaai.org/tools-for-the-public/conditions-library/allergies/medications-and-older-adults] (American Academy of Allergy, Asthma, and Immunology)

Patient Handouts

- Allergies, asthma, and dust [https://medlineplus.gov/ency/patientinstructions/000487.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000487.htm]

- Allergies, asthma, and molds [https://medlineplus.gov/ency/patientinstructions/000488.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000488.htm]
- Allergies, asthma, and pollen [https://medlineplus.gov/ency/patientinstructions/000489.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000489.htm]
- Asthma [https://medlineplus.gov/ency/article/000141.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/article/000141.htm]
- Asthma - control drugs [https://medlineplus.gov/ency/patientinstructions/000005.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000005.htm]
- Asthma - quick-relief drugs [https://medlineplus.gov/ency/patientinstructions/000008.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000008.htm]
- Exercise-induced asthma [https://medlineplus.gov/ency/patientinstructions/000036.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000036.htm]
- How to breathe when you are short of breath [https://medlineplus.gov/ency/patientinstructions/000053.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000053.htm]
- How to use a nebulizer [https://medlineplus.gov/ency/patientinstructions/000006.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000006.htm]
- How to use an inhaler - no spacer [https://medlineplus.gov/ency/patientinstructions/000041.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000041.htm]
- How to use an inhaler - with spacer [https://medlineplus.gov/ency/patientinstructions/000042.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000042.htm]
- Pulmonary function tests [https://medlineplus.gov/ency/article/003853.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/article/003853.htm]
- Signs of an asthma attack [https://medlineplus.gov/ency/patientinstructions/000062.htm] (Medical Encyclopedia)
  Also in Spanish [https://medlineplus.gov/spanish/ency/patientinstructions/000062.htm]

MEDICAL ENCYCLOPEDIA

Allergies, asthma, and dust [https://medlineplus.gov/ency/patientinstructions/000487.htm]

Allergies, asthma, and molds [https://medlineplus.gov/ency/patientinstructions/000488.htm]

Allergies, asthma, and pollen [https://medlineplus.gov/ency/patientinstructions/000489.htm]

Asthma [https://medlineplus.gov/ency/article/000141.htm]

Asthma - control drugs [https://medlineplus.gov/ency/patientinstructions/000005.htm]

Asthma - quick-relief drugs [https://medlineplus.gov/ency/patientinstructions/000008.htm]

Asthma in adults - what to ask the doctor [https://medlineplus.gov/ency/patientinstructions/000208.htm]

Exercise-induced asthma [https://medlineplus.gov/ency/patientinstructions/000036.htm]

How to breathe when you are short of breath [https://medlineplus.gov/ency/patientinstructions/000053.htm]

How to use a nebulizer [https://medlineplus.gov/ency/patientinstructions/000006.htm]

How to use an inhaler - no spacer [https://medlineplus.gov/ency/patientinstructions/000041.htm]

How to use an inhaler - with spacer [https://medlineplus.gov/ency/patientinstructions/000042.htm]

How to use your peak flow meter [https://medlineplus.gov/ency/patientinstructions/000043.htm]

Make peak flow a habit [https://medlineplus.gov/ency/patientinstructions/000046.htm]

Metered dose inhaler use - Series [https://medlineplus.gov/ency/presentations/100200_1.htm]

Nebulizer use - series [https://medlineplus.gov/ency/presentations/100201_1.htm]

Occupational asthma [https://medlineplus.gov/ency/article/000110.htm]

Peak flow meter use - Series [https://medlineplus.gov/ency/presentations/100202_1.htm]

Pulmonary function tests [https://medlineplus.gov/ency/article/003853.htm]

Signs of an asthma attack [https://medlineplus.gov/ency/patientinstructions/000062.htm]

Smoking and asthma [https://medlineplus.gov/ency/patientinstructions/000504.htm]

Spacer use - Series [https://medlineplus.gov/ency/presentations/100203_1.htm]

Stay away from asthma triggers [https://medlineplus.gov/ency/patientinstructions/000064.htm]

Traveling with breathing problems [https://medlineplus.gov/ency/patientinstructions/000066.htm]

Wheezing [https://medlineplus.gov/ency/article/003070.htm]

Related Health Topics

Air Pollution [https://medlineplus.gov/airpollution.html]

Allergy [https://medlineplus.gov/allergy.html]

Asthma in Children [https://medlineplus.gov/asthmainchildren.html]

Breathing Problems [https://medlineplus.gov/breathingproblems.html]

Indoor Air Pollution [https://medlineplus.gov/indoorairpollution.html]

National Institutes of Health

The primary NIH organization for research on *Asthma* is the National Heart, Lung, and Blood Institute [http://www.nhlbi.nih.gov/]

Other Languages

Find health information in languages other than English [https://medlineplus.gov/languages/asthma.html] on *Asthma*

NIH MedlinePlus Magazine

Breathing Like Everyone Else [https://magazine.medlineplus.gov/article/breathing-like-everyone-else]

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